

 B.TECH AIML MAJOR PROJECT

SentinelPdM

Offline Predictive Maintenance for Industrial Rotating Machinery

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INSTITUTION

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Edge PWA vs Legacy Cloud

SentinelPdM (Edge PWA)

100% OFFLINE

Network Dependency	Zero (Air-Gapped)
Inference Latency	< 0.88 ms (In-Memory)
Hosting & Ingress Costs	\$0 / Month
Data Sovereignty	Never leaves plant

Legacy Enterprise Cloud

INTERNET DEPENDENT

Network Dependency	Constant Ingress Link
Inference Latency	250 - 800 ms (HTTP)
Hosting & Ingress Costs	High Cloud Bills
Data Sovereignty	Firewall Violations

The Industrial Dilemma

⚠️ **Unplanned Plant Outages**

\$100k+/Hour Loss

Rotating unit breakdown shuts down entire downstream continuous processes.

🗓️ **Wasteful Calendar Routines**

70%+ False Overhauls

Preventive overhauls open healthy machines, introducing human re-assembly flaws.

🔒 **Air-Gapped Network Wall**

OT Cyber Policy

SCADA & historian firewalls strictly prohibit outbound telemetry into public clouds.



5 Monitored Machine Classes



Centrifugal Pump

Cavitation, mechanical seal wear & impeller imbalance.

1,750 RPM



Gas Compressor

Valve leakage, ring degradation & thermal runaway.

1,450 RPM



Steam Turbine

Shaft unbalance, thermal casing expansion & blade wear.

3,200 RPM



Induction Motor

Stator winding breakdown, eccentric air gaps & bearings.

1,780 RPM



Bulk Conveyor

Roller bearing seizure, belt slip & motor drag overload.

900 RPM

6-Channel Condition Sensing



Vibration RMS
Kinetic Energy

mm/s

nv (+2.56)



Bearing Temp
Thermal Dissipation

°C

nt (+0.32)



Shaft Velocity
Rotational Speed

RPM

nr (+0.15)



Oil Health
Lubricant Film

0 - 100

no (+0.34)



Process Pressure
Fluid Compression

bar

np
(+0.25)



Motor Current
Electrical Load

Amperes

nc (+0.13)



Ultra-Lightweight Inference

STATIC ASSET SIZE

< 2 KB

Embedded model-weights.json bundle

INFERENCE LATENCY

0.88 ms

Sub-millisecond native JavaScript evaluation

Deterministic Mathematical Formulation

Zero external ML frameworks (No Python, TensorFlow, or ONNX runtimes needed).

$$z = b + \sum_{i=1}^6 w_i x_i, \quad p = \frac{1}{1 + e^{-z}}$$

OPTIMIZER

Batch GD + L2

CONVERGED LOSS

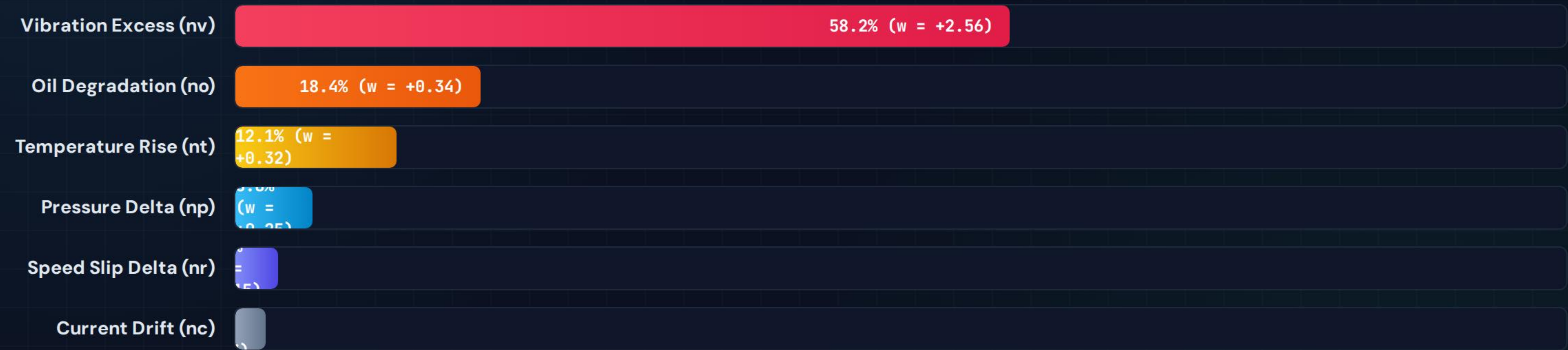
0.1437 BCE

Time-to-Failure Forecasting



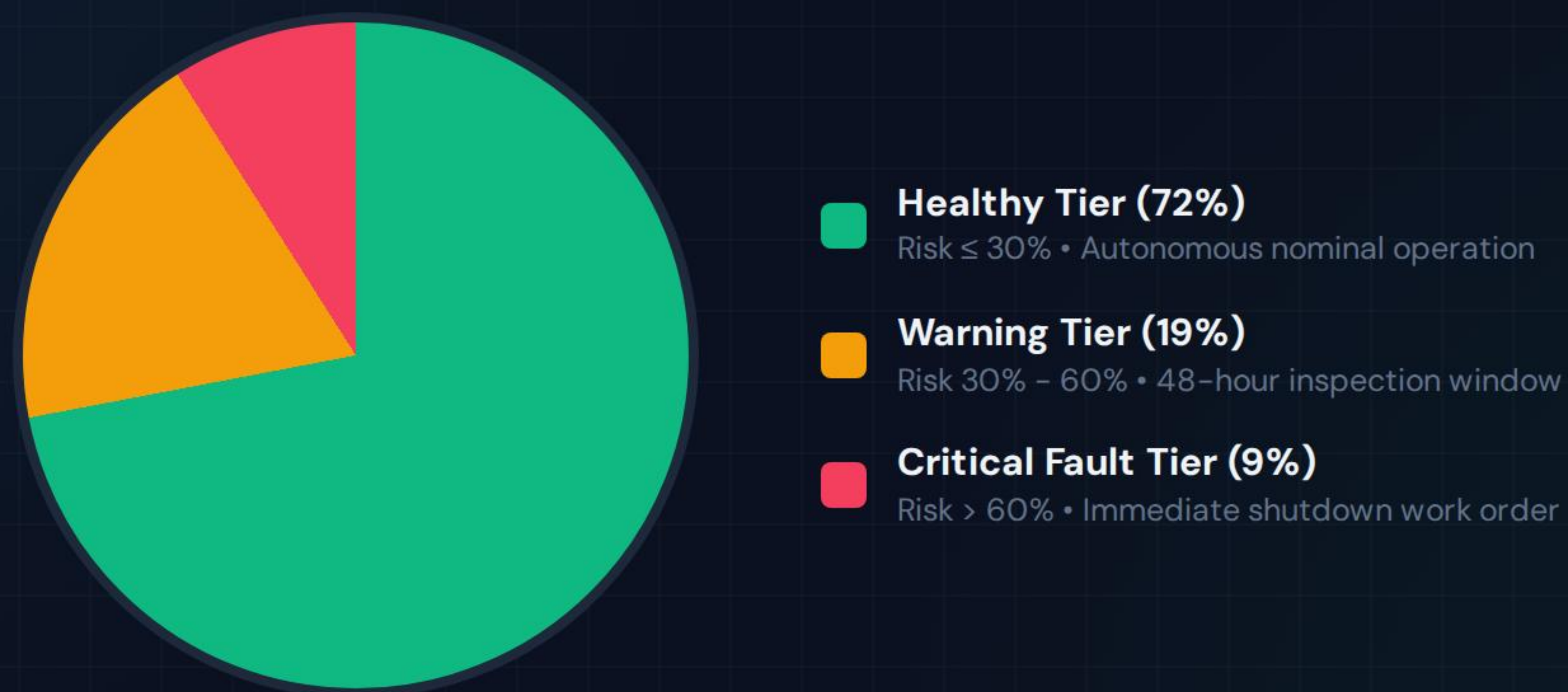
Log-linear extrapolation across rolling 48-hour risk histories calculates precise work order intervention windows.

Explainable AI Attribution



Exact mathematical feature attribution removes black-box doubt, highlighting physical fault culprits instantly.

Fleet Health Distribution



Real-time triage enables control-room operators to immediately isolate high-risk machinery.

Prescriptive Maintenance Tasks



Vibration Spike Trigger

CRITICAL (>60%)

Immediate controlled shutdown: Inspect rolling element bearings, coupling, and shaft alignment.



Oil Breakdown Trigger

CRITICAL (>60%)

Emergency lubricant flush: Clean sump chamber, replace oil filters, and verify particulate count.



Thermal Warning Trigger

WARNING (30-60%)

Inspect cooling water lines, examine heat exchangers, and replenish bearing grease.



Nominal Band Execution

HEALTHY (≤30%)

No action required: Continue continuous autonomous sensor condition logging.

System Benchmark Results

93.87%

Test Accuracy

40k Held-Out Samples

0.710

Precision Score

Low False Alarms

124 ms

Cold PWA Boot

Cache-First Service Worker

60 FPS

UI Rendering

14.1 ms SVG Frame Cycle

Train Accuracy (160k)

94.81%

Test Accuracy (40k)

93.87%

Minimal 0.94% train/test variance confirms L2 regularization effectively prevents model overfitting.

✓ IMPLEMENTATION COMPLETE • VERIFIED

Questions & Discussion

SentinelPdM demonstrates that production-grade predictive intelligence, forecasting, and explainability run reliably in air-gapped industrial edge environments.

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Image Sources



Thumbnail for <https://www.petrosync.com/blog/wp-content/uploads/2025/10/Rotating-Equipment-scaled.webp>
Source: www.petrosync.com



<https://www.phaseivengr.com/wp-content/uploads/2018/12/Motor-Monitoring-Rendering-with-Leap-parts-and-graph-inset-500x375.jpg>
Source: www.phaseivengr.com